

Linear regression: Abalone growth and pH example

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Set up

```
library(tidyverse) # general use
library(janitor) # cleaning data frames
library(here) # file/folder organization
library(readxl) # reading .xlsx files
library(ggeffects) # generating model predictions
library(flextable) # creating tables
```

```
# abalone data from Hamilton et al. 2022
abalone <- read_xlsx(here("data", "Abalone IMTA_growth and pH.xlsx"))
```

Abalone example

Data from Hamilton et al. 2022. “Integrated multi-trophic aquaculture mitigates the effects of ocean acidification: Seaweeds raise system pH and improve growth of juvenile abalone.”
<https://doi.org/10.1016/j.aquaculture.2022.738571>

a. Questions and hypotheses

Question: How does pH predict abalone growth (measured in change in shell surface area per day, $\text{mm}^2 \text{d}^{-1}$)?

H_0 : pH does not predict abalone growth.

H_A : pH has an effect on abalone growth.

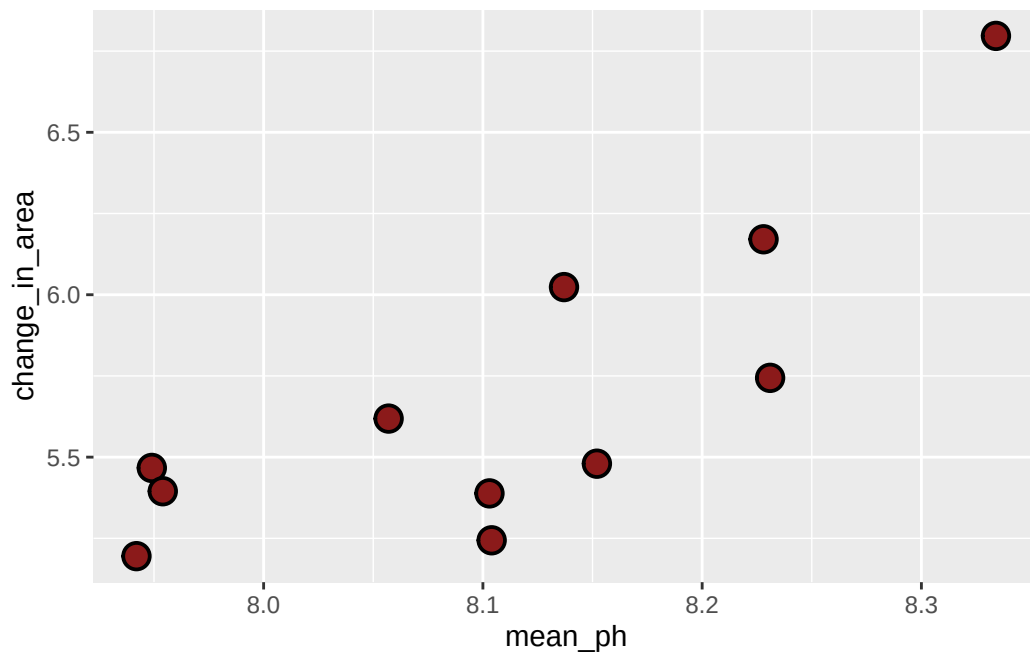
b. Cleaning

```
# creating new object from abalone dataframe
abalone_clean <- abalone |>
  # cleaning column names
  clean_names() |>
  # selecting columns of interest
  select(mean_p_h, change_in_area_mm2_d_1_25) |>
  # renaming columns for clarity
  rename(mean_ph = mean_p_h,
         change_in_area = change_in_area_mm2_d_1_25)
```

Don't forget to look at your data! Use `View(abalone_clean)` in the Console.

c. Exploratory data visualization

```
# base layer: ggplot
ggplot(data = abalone_clean, # using abalone_clean data
       aes(x = mean_ph, # x-axis: mean pH, y-axis: shell surface area change
           y = change_in_area)) +
# first layer: geom_point to represent individual observation
# changing size, shape, color
geom_point(size = 4,
           stroke = 1,
           fill = "firebrick4",
           shape = 21)
```



Where does the missing value come from?

The missing value comes from a pair of tanks in which the pH sensor malfunctioned, so there is no pH measurement for those tanks.

Don't forget to turn your warnings off either in the individual code chunk or in the YAML at the top of the file.

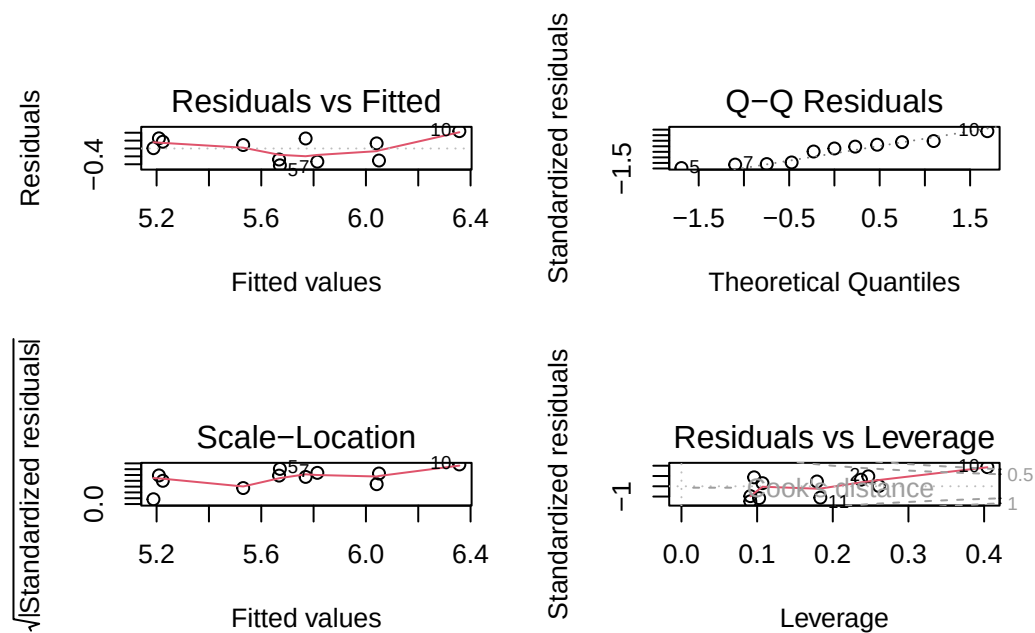
d. Abalone model

Model fitting

```
abalone_model <- lm(  
  change_in_area ~ mean_ph, # formula: change in shell SA as a function of pH  
  data = abalone_clean      # data from: clean abalone data  
)
```

Diagnostics

```
par(mfrow = c(2, 2)) # displays plots in a 2x2 grid  
plot(abalone_model)  # display diagnostic plots
```



Model summary

```
# more information about the model  
summary(abalone_model)
```

```

Call:
lm(formula = change_in_area ~ mean_ph, data = abalone_clean)

Residuals:
    Min       1Q   Median       3Q      Max
-0.42714 -0.29282  0.08769  0.21258  0.43965

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -18.5010     6.1601  -3.003  0.01488 *
mean_ph      2.9827     0.7596   3.926  0.00348 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3063 on 9 degrees of freedom
(1 observation deleted due to missingness)
Multiple R-squared:  0.6314,    Adjusted R-squared:  0.5905
F-statistic: 15.42 on 1 and 9 DF,  p-value: 0.003477

```

What is the slope estimate (with standard error?)

$$\beta_1 = 2.98 \pm 0.76$$

What is the intercept estimate (with standard error?)

$$\beta_0 = -18.50, \pm = 6.16$$

Generating predictions

```

# creating a new object called abalone_preds
abalone_preds <- ggpredict(
  abalone_model,      # model object
  terms = "mean_ph"   # predictor (in quotation marks)
)

# display the predictions
abalone_preds

```

Predicted values of change_in_area

```
mean_ph | Predicted |      95% CI
```

```
-----
7.94 |      5.19 | 4.83, 5.54
7.95 |      5.21 | 4.86, 5.55
8.06 |      5.53 | 5.30, 5.76
8.10 |      5.67 | 5.46, 5.88
8.10 |      5.67 | 5.46, 5.88
8.14 |      5.77 | 5.55, 5.98
8.15 |      5.81 | 5.59, 6.04
8.33 |      6.36 | 5.92, 6.80
```

Look at the column names using `colnames(abalone_preds)` in the Console.

What are the column names?

`x`, `predicted`, `std.error`, `conf.low`, `conf.high`, `group` (not worrying about `group` for now)

Look at the actual object (by clicking on it in Environment).

Look at the class of the object using `class(abalone_preds)` in the Console.

What is this type of object?

This is a ggeffects object, but it is also a data frame

View `abalone_preds` like a data frame.

Why generate model predictions?

This dataset does not include any actual observations of abalone growth at $\text{pH} = 8$.

However, the model can tell us a prediction for what abalone growth could be if $\text{pH} = 8$ based on the existing data.

Figure out what abalone growth the model would predict at $\text{pH} = 8$.

```
# finding the model prediction at a specific value
ggpredict(
  abalone_model,      # model object
  terms = "mean_ph[8]" # predictor (in quotation marks) and predictor value in brackets
)
```

Predicted values of change_in_area

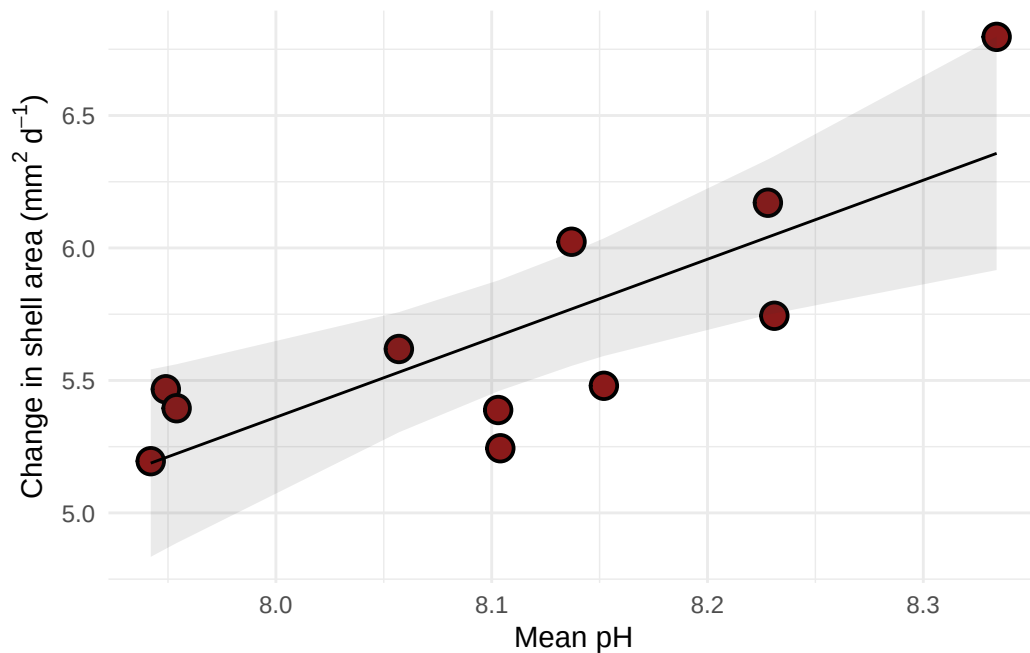
```
mean_ph | Predicted |      95% CI
-----
      8 |      5.36 | 5.08, 5.64
```

How would you interpret this?

At pH = 8, the model predicts abalone shell growth of is 5.36 mm per day [95% CI = 5.08, 5.64].

Visualizing model predictions and data

```
# base layer: ggplot
# using clean data frame
ggplot(data = abalone_clean,
       aes(x = mean_ph,
           y = change_in_area)) +
# first layer: points representing abalones
geom_point(size = 4,
           stroke = 1,
           fill = "firebrick4",
           shape = 21) +
# second layer: ribbon representing confidence interval
# using predictions data frame
geom_ribbon(data = abalone_preds,
          aes(x = x,
              y = predicted,
              ymin = conf.low,
              ymax = conf.high),
          alpha = 0.1) +
# third layer: line representing model predictions
# using predictions data frame
geom_line(data = abalone_preds,
         aes(x = x,
             y = predicted)) +
# axis labels
labs(x = "Mean pH",
     y = expression("Change in shell area (*mm^{2}~d^{-1}*")) +
# theme
theme_minimal()
```



Creating a table with model coefficients, 95% confidence intervals, and more

Note: this function is from `flextable`.

```
as_flextable(abalone_model)
```

	Estimate	Standard Error	t value	Pr(> t)
(Intercept)	-18.501	6.160	-3.003	0.0149 *
mean_ph	2.983	0.760	3.926	0.0035 **

*Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05*

Residual standard error: 0.3063 on 9 degrees of freedom

Multiple R-squared: 0.6314, Adjusted R-squared: 0.5905

F-statistic: 15.42 on 9 and 1 DF, p-value: 0.0035

END OF ABALONE EXAMPLE